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ORIGINAL PAPER

Interpretable machine learning framework for air quality prediction in Istanbul using Shapley additive explanations (SHAP)

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Abstract

This study develops a season-aware machine-learning (ML) framework to predict hourly concentrations of PM_{10}, $PM_{2.5}$ and O_3 across İstanbul. A comprehensive 2021–2023 dataset was compiled from three co-located air-quality and meteorological monitoring stations that typify contrasting source regimes, i.e., a traffic-dominated urban site, a rural background site, and a semi-urban coastal site. Seven ML algorithms, namely eXtreme Gradient Boosting (XGBoost), Extra Trees (ETR), Random Forest (RF), Adaptive Boosting (AdaBoost), Multi-Layer Perceptron (MLP), k-Nearest Neighbors (KNN) and Support Vector Regression (SVR), were utilized to establish a holistic comparison scheme. Hyperparameters were optimized using five-fold cross-validated Bayesian search, and models were evaluated with various performance indicators on season-withheld test sets. In the winter months, ETR achieved a mean $R^2=0.93$ (RMSE ≈ 10 μg/m^3) for PM_{10} at Bağcılar, while XGBoost yielded $R^2=0.88$ for O_3 at the same site. Summer predictions were more challenging. PM_{10} skill in rural Arnavutköy dropped to $R^2=0.61$ despite strong training fits, highlighting over-fitting risks under complex, non-stationary chemical conditions. By contrast, MLP maintained robust urban performance for $PM_{2.5}$ (summer test $R^2=0.80$) and KNN provided the most stable O_3 prediction in rural areas ($R^2=0.74$). To enhance interpretability, SHAP (SHapley Additive exPlanations) analysis was applied to the best-performing models, enabling a transparent assessment of how meteorological and co-pollutant inputs shaped predictions at each site. The proposed framework demonstrates that data-driven models can complement traditional air-quality modeling systems by providing station-level insights and interpretable relationships between pollutants and meteorological drivers, supporting air-quality assessment and policy-relevant analyses in rapidly urbanizing regions.

Keywords Machine learning · İstanbul · PM_{10} · $PM_{2.5}$ · O_3 · SHAP

1 Introduction

With increasing urbanization and population growth, air pollution has become one of the most important problems today. With the rapid increase in urban population, the number of people living in urban areas will be 68% of the world's population in 2050 (DESA 2024). According to the World Health Organization (WHO), it is estimated that there were approximately 3.7 million premature deaths due to air pollution in 2012 (WHO 2022). Especially in global studies and among the widely used air pollutants, air quality standards, including limit values under the U.S. EPA's National Ambient Air Quality Standards (NAAQS), have been established for particles smaller than 10 µm in diameter (PM_{10}) and particles smaller than 2.5 µm in diameter ($PM_{2.5}$) as well as nitrogen dioxide (NO_2) to protect the environment and human health (Suh et al. 2000). While the health and environmental impacts of air pollution are well established, improving estimation of pollutant concentrations, particularly under data gaps or monitoring limitations, enables more robust analysis and actionable insights.

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In air pollution prediction, chemical models, statistical and machine learning methods are used. Chemistry-transport models (CTMs) use first principles to solve mathematical equations. The most widely used among CTMs models are the Community Multiscale Air Quality (CMAQ) model (Byun and Ching 1999; Chen et al. 2014) and the Weather Research and Forecasting model coupled to Chemistry (WRF-Chem) (Grell et al. 2005), as well as other systems such as the Comprehensive Air Quality Model with Extensions (CAMx) (Shao et al. 2025) and LOTOS-EUROS (Hinestroza-Ramirez et al. 2023), which are widely applied in regional air quality studies. However, simulations of these models are expensive and time consuming since they involve chemical processes (Zhang et al. 2011). CTMs simulate pollutant transport and chemistry in three dimensions, whereas data-driven methods typically operate on point-based monitoring data; therefore, the two approaches are complementary. At the same time, statistical methods are inadequate in solving nonlinear problems and have difficulty in estimating associated problems due to sophisticated atmospheric systems exhibiting nonlinear oscillations (Lorenz 1963). In statistical methods, especially multiple linear regression models (MLR) (Gupta and Christopher 2009) and autoregressive integrated moving average (ARIMA) model (Kumar and Jain 2010) were frequently used. However, the use of ML techniques has given good results in the prediction of air pollutants recently. For example, nonlinear models such as support vector machine (SVM), multi-layer perceptron (MLP), vector auto-regressive moving-average (VARMA) were developed to predict PM_{10} concentration in Spain (Nieto et al. 2018). In another study, the Support Vector Regression (SVR) algorithm was used to predict the air quality of pollutants in California (Castelli et al. 2020). PM_{10} concentrations were estimated using ML prediction models. The results showed that the actual values and the predicted values were very close to each other in Taiwan (Doreswamy et al. 2020). ML techniques have become a crucial tool in predicting pollutant concentrations due to their ability to deal with nonlinear problems (Zhu et al. 2021, 2022). In addition, meteorological variables are significant in air quality prediction, as they govern dispersion (wind speed/direction), photochemical reactions (temperature, solar radiation), and boundary layer dynamics (mixing height), all of which strongly shape pollutant concentrations (Seinfeld and Pandis 2016).

Meteorological variables such as wind speed and direction are some of the most important variables for prediction (Madan et al. 2020). For example, $PM_{2.5}$ was estimated using Wavelet Neural Network (WNN) technique together with Ozone (O_3), Carbon Monoxide (CO), NO_2, Sulfur dioxide (SO_2), PM_{10}, $PM_{2.5}$, temperature and relative humidity (Zhang et al. 2018). MLR, ARIMA, Classification and regression tree, and neural networks (NN) were used to predict urban haze by means of different meteorological variables (e.g., temperature, humidity and atmospheric pressure) values (Lu et al. 2018). ML provided more accurate outcomes compared to the feedforward NN techniques with backpropagation (FFANN-BP) and MLR techniques (Zhang and Ding 2017). Artificial neural network (ANN) and logistic regression models are also widely used in air quality prediction (Patil et al. 2020). ANN, boosted regression trees (BRT) and SVM models were used to evaluate the reduction process of PM_{10} and $PM_{2.5}$, and 95% success was achieved for PM_{10} in all model results (Suleiman et al. 2019). In addition, PM_{10} concentrations together with meteorological variables such as temperature, precipitation and wind speed were estimated using Bayesian network (BN), SVM and decision tree (DT) technology (Mehdipour et al. 2018). Many similar methods have widely been used in the literature. In particular, Random Forest (RF) and extreme gradient boosting (XGBoost) (Kamińska , 2018; Wen et al. 2019; Choubin et al. 2020) for the temporal and spatial prediction of $PM_{2.5}$ area and for the prediction of PM_{10} expectation-Nearest Neighbor Regression (KNNR), SVM, SVR, bosted regression tree (BRT), RF, XGBoost (Plocoste and Laventure 2023; Bozdağ et al. 2020; Shaziayani et al. 2022; Peng et al. 2022) methods have widely been acknowledged by the research community.

Some of the ML-based studies have been conducted for air quality prediction in İstanbul. For example, a stacking ensemble machine learning approach was used for $PM_{2.5}$ prediction (Emeç and Yurtsever 2024). In another study, discrete wavelet transforms (DWT) and deep learning method long short-term memory (LSTM) models were used for PM_{10} prediction (Taşağıl Arslan and Kılıç Depren 2024). Similarly, some pollutants such as PM_{10} and O_3 were predicted using Hybrid CNN + LSTM deep neural network (Gilik et al. 2022). Also, LTSM, Recurrent Neural Network (RNN), and MLP models were used for PM_{10} and SO_2 prediction. LTSM results were found to give better results than MLP and RNN for PM_{10} and SO_2 (Das et al. 2022). Eren et al. (2023) predicted $PM_{2.5}$ using long-term air pollution data with LSTM, RNN, and GRU models. In contrast, the present study expands on this by benchmarking seven different ML algorithms across multiple pollutants (PM_{10}, $PM_{2.5}$ and O_3), explicitly considering seasonal regimes and site typologies, and further incorporating SHAP analysis for model interpretability. Moreover, The SHAP method is also used to show what the value to be estimated depends on and how these values affect it.

Recent research has applied ML to particulate matter and ozone. For instance, Borlaza et al. (2021) combined MLR with MLP to apportion PM_{10} oxidative potential sources, highlighting traffic and biomass burning as dominant contributors. During the COVID-19 lockdown, Borlaza et al. (2023) used RF models to estimate business-as-usual oxidative potential and confirmed the strong role of combustion sources at both urban and traffic sites. Extending source apportionment approaches, Borlaza-Lacoste et al. (2024) applied a dispersion-normalized PMF method in New York City to track long-term VOC source contributions, demonstrating their relevance for ozone exceedances. Beyond particulate matter (PM), Grange and Carslaw (2019) introduced meteorological normalization with Random Forest to robustly identify intervention impacts in air quality time series. Hou et al. (2022) coupled RF with SHAP to interpret the meteorological and photochemical drivers of haze episodes in China, while Xu et al. (2023) used ML and structure mining to visualize the interactions among multiple drivers of ozone formation, particularly highlighting the role of isoprene–NO_x dynamics. Collectively, these studies illustrate the power of ML for characterizing pollutant dynamics, but they often focus on specific contexts, individual pollutants, or limited site types. Building on this body of work, the present study systematically compares seven ML algorithms for predicting PM_{10}, $PM_{2.5}$ and O_3 across Istanbul, explicitly considering seasonality and site typologies, while extending interpretability through a comprehensive SHAP analysis.

In this study, air pollution estimates were performed using air pollution data (PM_{10}, $PM_{2.5}$ and O_3) and meteorological data (Solar radiation (W/m^2), air pressure (hPa), air temperature (°C), cabin temperature (°C), relative humidity (%), wind speed (m/s), wind direction (°) and rainfall (mm) for the summer and winter periods between 2021 and 2023. All models were trained and evaluated on hourly data. For each AQMS, the summer periods include June, July, and August of 2021, 2022, and 2023, resulting in a total of 6,624 hourly measurements. The winter periods include January–February 2021, December 2021–February 2022, December 2022–February 2023, and December 2023, resulting in a total of 6,480 hourly measurements. This high-resolution coverage ensures robust sample sizes for model benchmarking. Previous studies used long-term data sets and without distinction between summer and winter (Emeç and Yurtsever 2024; Gilik et al. 2022; Yang et al. 2021), chose a single region, such as an urban area (Rahimpour et al. 2021; Delavar et al. 2019; Mampitiya et al., 2023), LSTM, DWT, RF, XGboost methods (Zhang et al. 2023a, b; Muley et al. 2023; Liu et al. 2023) and with a long distance between meteorological stations and AQMS stations (Mao et al. 2021; Ma et al. 2020). On the other hand, this study took summer and winter months into account separately, stations were selected from urban, urban-traffic, rural and urban background regions, meteorological and pollutant data were measured from the same station and divergent approaches (including XGBoost, KNN, ETR, RF, AdaBoost, SVR, MLP) from different ML families were employed. Accordingly, the major research objectives of this study are five-fold: (i) although the previous studies have examined the potential of data-driven algorithms in predicting only one pollutant, this study devoted significant efforts to the estimation of multiple pollutants (i.e., PM_{10}, $PM_{2.5}$ and O_3) to holistically analyze the success of the ML approaches, (ii) The impacts of divergent geographical regions, i.e., residential, rural, semi-rural, on the air quality were considered (iii) The influence of seasonal variations on model accuracy by explicitly comparing summer and winter conditions in Istanbul was assessed (iv) Despite the existing body of knowledge taking the pollutant concentrations and meteorological inputs into account from different stations, the current research incorporated the records regarding both pollutant concentrations and the meteorological forcing into the dataset from the same station, inhibiting a potentially biased estimations due to locational variations (v) Apart from the literature that generally focused on the utilization of XGBoost, KNN, ExtraTrees approaches, the present work incorporated the SHAP technique into the predictive framework to further support the pertinent literature and to provide model interpretability by quantifying the contribution of each predictor to the model output.

2 Research area and data

Istanbul is the largest city in Türkiye in terms of population and has approximately 16 million people living in the city (TUIK 2024). The city is divided into two by the Bosphorus Strait between Europe and Asia, and the Black Sea is in the north and the Marmara Sea is closed to the south. In this study, three Air Quality Monitoring Stations (AQMS) were selected based on geographical region and urbanization. The characteristics and names of these three stations were selected as Urban (Bağcılar AQMS), rural (Arnavutköy AQMS) and urban background (Şile AQMS), respectively. The location of İstanbul region and the location of AQMSs are shown in Fig. 1.

In the current research, air quality data (PM_{10}, $PM_{2.5}$ and O_3) and meteorological data (solar radiation, air pressure, air temperature, cabin temperature, humidity, wind speed and direction, rain) were obtained from AQMS stations. All data obtained started at 00:00 on 01.01.2021 and ended at 23:00 on 31.12.2023. Then, these data were divided into summer months (June, July and August) and winter months (December, January and February). Before applying the ML models, the datasets were diligently investigated. After initial cleaning, the raw data was already validated by the İstanbul Metropolitan Municipality. Also, we removed implausible or instrument-related artifacts (e.g., negative

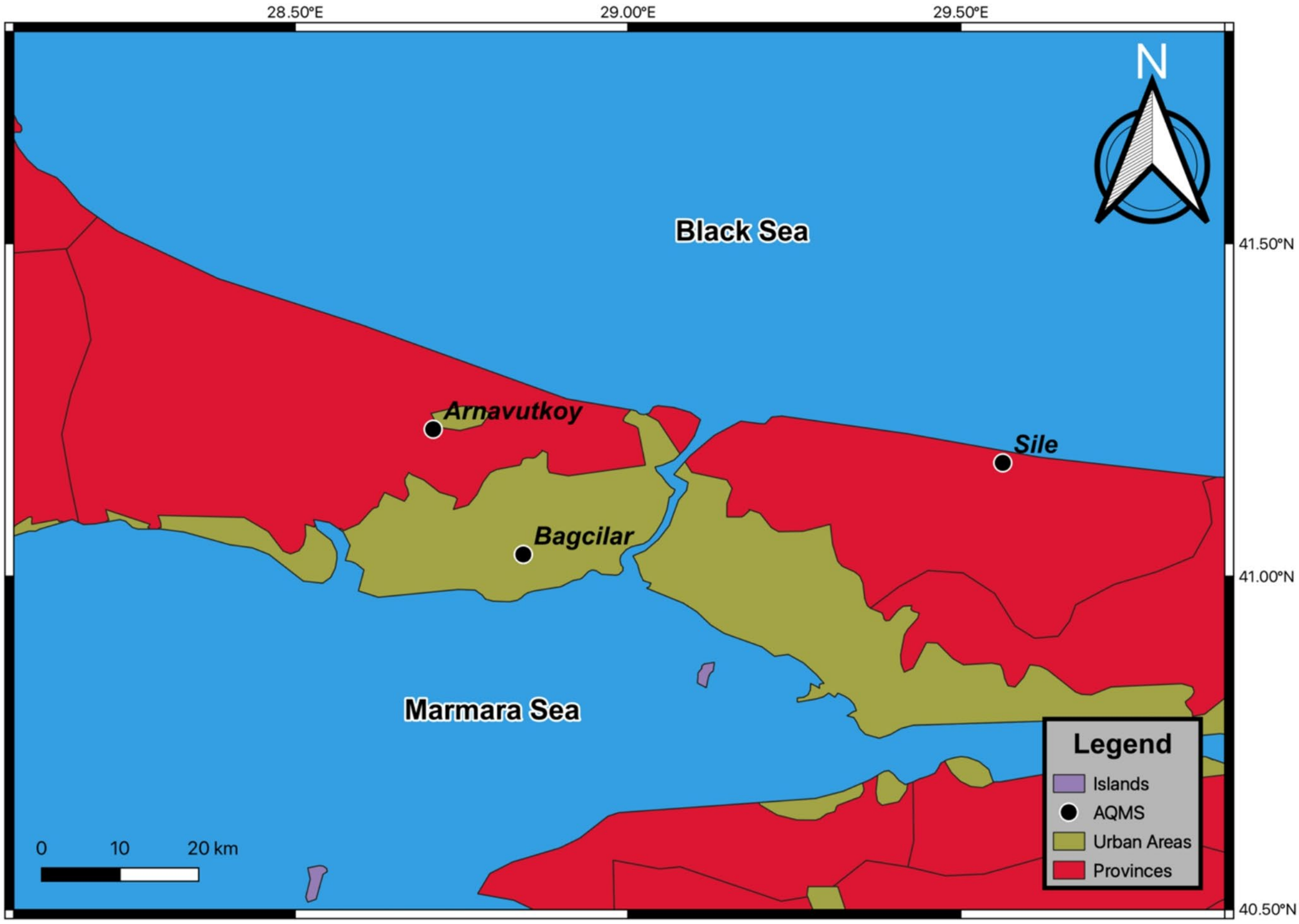

Fig. 1 Location of İstanbul and AQMS's

values, flat-line sequences, unrealistic maxima), while retaining genuine extreme pollution episodes such as dust transport events. Prior to applying the algorithms, we also performed descriptive analyses, including summary statistics and exploratory time-series plots, to characterize the dataset and ensure robustness for subsequent predictions. The data sets were subjected to a missing value imputation. Table 1 shows the missing data information before filling the data. Also, the complete mapping of variables to imputation methods is provided in supplementary material (Table S1 and S2).

Table 1 shows the number of data that should be in winter and summer and the number of data that is available. Before filling in missing data, the distribution of the data was examined and different techniques were used for each data. Many data imputation techniques are used in studies conducted in the relevant literature for different regions across the globe. For example, in a study conducted in China, meteorological data were filled with the linear regression method (Qin et al. 2021). In addition, in another study focusing on the analysis of air quality, the simple average method is frequently used in studies with small data sets and when missing data is low (Quinteros et al. 2019). Similarly, forward and backward imputation techniques were also used to fill in missing data (Arnaut et al. 2024). Since meteorological data are generally time series and are inherently continuous, the linear interpolation method was applied. However, in some data, when they form the normal distribution, the arithmetic mean method was used. Air quality data is more complete than meteorological data and the number of missing data is quite low. In addition, Forward and Backward imputation techniques were used for some missing data due to the time sequence. Tables 2 and 3 show the descriptive statistics calculated separately for each station in the summer and winter months after the data were filled in, respectively.

As seen in Tables 2 and 3, the average values of PM concentrations are lower in summer than in winter, consistent with previous findings in Istanbul (Birinci et al. 2023a). Skewness and kurtosis are descriptive statistics commonly used to characterize data distributions. Skewness indicates whether the distribution is symmetric or skewed, while kurtosis reflects the weight of the tails and the presence of extreme values. In our dataset, several variables, including PM_{10}, $PM_{2.5}$, and rainfall that exhibit strongly

Table 1 Missing data summary of the dataset used

Seasons	Variables	Missing Hours per AQMS			
		Expected Hourly Records	Arnavutköy	Bağcılar	Şile
WINTER	Solar Radiation (W/m^2)	6480	3	4	-
	Air Pressure (hPa)	6480	3	4	224
	Air Temperature (°C)	6480	6	13	224
	Relative Humidity (%)	6480	103	4	224
	Wind Speed (m/sn)	6480	21	16	224
	Wind Direction (°)	6480	43	1107	224
	Rainfall (mm)	6480	0	3	-
	NO_2 ($\mu g/m^3$)	6480	17	171	340
	NO_X ($\mu g/m^3$)	6480	20	155	348
	O_3 ($\mu g/m^3$)	6480	568	12	306
	PM_{10} ($\mu g/m^3$)	6480	119	99	420
	$PM_{2.5}$ ($\mu g/m^3$)	6480	225	81	-
SUMMER	Solar Radiation (W/m^2)	6624	1	84	-
	Air Pressure (hPa)	6624	1	84	117
	Air Temperature (°C)	6624	6	86	117
	Relative Humidity (%)	6624	20	84	117
	Wind Speed (m/sn)	6624	1	84	117
	Wind Direction (°)	6624	8	257	118
	Rainfall (mm)	6624	0	83	126
	NO_2 ($\mu g/m^3$)	6624	6	211	310
	NO_X ($\mu g/m^3$)	6624	127	212	316
	O_3 ($\mu g/m^3$)	6624	763	377	177
	PM_{10} ($\mu g/m^3$)	6624	181	131	197
	$PM_{2.5}$ ($\mu g/m^3$)	6624	387	117	-

*Expected Hourly Records indicate the total number of hours within each seasonal block (6,480 for winter, 6,624 for summer). Missing Hours refer to the number of individual hourly observations not recorded at each site. Missing data mainly arose from instrument downtime, communication failures, or quality control exclusions. "–" indicates that the variable was not measured at the corresponding site.

positive skewness and extremely high kurtosis, particularly for rainfall (K > 1000). These values clearly show that the data depart from normality and are dominated by rare but extreme episodes (e.g., heavy rainfall, dust transport, or pollution spikes). While air quality variables are typically log-normally distributed and precipitation is episodic by nature, reporting skewness and kurtosis here provides a descriptive characterization of the dataset and supports the rationale for using imputation methods and ML models that do not assume normality.

3 Materials and methods

For model training, the dataset was partitioned by season, with two summers (JJA 2021–2023) and two winters (DJF 2021–2023) used for analysis. This seasonal split avoided temporal leakage and allowed the models to be evaluated under distinct meteorological conditions. While month or site could be treated as categorical predictors, we chose season- and site-specific models to better capture distinct meteorological regimes and source environments, improving interpretability and reducing heterogeneity. To avoid time-series leakage, models were trained and evaluated using chronological (blocked) splits within each season (summers and winters analyzed separately), keeping the temporal order intact and reserving a strictly held-out test block for final evaluation.

We compared many ML approaches (i.e., XGBoost, Extra Trees, RF, AdaBoost, Gradient Boosting, KNN, MLP, and SVR) to achieve the model ensuring the most accurate air quality outcomes. Hyperparameter optimization was carried out using the Bayesian search separately for each algorithm using a systematic search procedure that evaluates candidate configurations on a validation subset and selects the one minimizing validation error; the chosen configuration was then refit on the training data and assessed on the held-out test set. For each algorithm, relevant hyperparameters (e.g., depth and number of trees for ensembles, kernel and penalty terms for SVR, network architecture for MLP) were tuned separately for each station and pollutant. To further avoid overfitting, we employed chronological blocked splits to prevent temporal leakage, applied regularization parameters in tree-based models (e.g., max depth, L1/L2 penalties in XGBoost), and dropout/weight decay in neural networks. The parameter spaces, tuning routine, and selected, alongside the evaluation metrics we report (e.g., RMSE, MAE, R^2, NSE, KGE, Willmott's d1) were all implemented in Python 3.10.

Table 2 Descriptive statistics for three AQMS in summer months (June, July and August)

Arnavuktöy	Mean	std	Min	25%	50%	75%	Max	Skewness	Kurtosis
Solar Radiation (W/m^2)	239	295.47	0	0	97.35	426.9	990.1	1	-0.42
Air Pressure (hPa)	996.38	3.47	984.7	994	996.5	998.8	1006.3	-0.19	0.04
Temperature (°C)	22.97	3.38	10.2	20.9	23.2	25	38.4	0.11	0.84
Relative Humidity (%)	76.4	14.67	20.6	67.08	78.1	88.4	100	-0.68	0.07
Wind Speed (m/sn)	3.43	1.51	0.2	2.2	3.3	4.4	9.3	0.41	-0.25
Wind Direction (°)	108.46	82.54	0	60.8	86.6	106.9	359.9	1.59	1.64
Rainfall (mm)	0.01	0.18	0	0	0	0	8	35.19	1387.71
NO_2 ($\mu g/m^3$)	17.63	13.41	0.7	9.4	13.8	21.1	148	2.51	9.77
NO_X ($\mu g/m^3$)	27.53	17.2	2.7	16.2	23.4	34.63	186.9	2.07	7.61
O_3 ($\mu g/m^3$)	61.52	18.7	2.1	50.8	61.1	72.9	163.2	0.07	1.14
PM_{10} ($\mu g/m^3$)	35.6	25.83	1	20	28.65	42	360.3	2.8	13.67
$PM_{2.5}$ ($\mu g/m^3$)	12.62	7.71	0.2	8	11	15.1	101.6	2.8	14.87
Bağcılar	Mean	std	Min	25%	50%	75%	Max	Skewness	Kurtosis
Solar Radiation (W/m^2)	241.37	287.81	0	0	80	477.53	929.3	0.82	-0.81
Air Pressure (hPa)	1008.37	3.53	995.6	1006.1	1008.4	1010.8	1018.2	-0.19	0.12
Air Temperature (°C)	25.17	3.67	12.8	22.9	25.1	27.5	39.1	0.06	0.24
Cabin Temperature (°C)	25.77	4.14	15.2	23.2	27.8	28.7	29.9	-1.01	-0.53
Relative Humidity (%)	66.28	15.03	21.9	55.5	67.6	79.1	98.3	-0.41	-0.64
Wind Speed (m/sn)	1.63	1.05	0	0.9	1.4	2.1	6.7	1.13	1.09
Wind Direction (°)	202.64	112.71	0	126.68	142.1	341.1	360	0.21	-1.32
Rainfall (mm)	0.28	1.28	0	0	0	0	36.6	20.64	481.94
NO_2 ($\mu g/m^3$)	32.25	24.34	2.4	15.8	25.2	38.9	215.9	2.06	5.62
NO_X ($\mu g/m^3$)	43.13	46.72	3	19	29.3	44.7	462.5	3.74	18.49
O_3 ($\mu g/m^3$)	54.91	25.56	0.8	38	54.91	74.4	186.6	-0.03	-0.1
PM_{10} ($\mu g/m^3$)	32.37	18.03	1.5	21	29	38	198.2	2.58	12.17
$PM_{2.5}$ ($\mu g/m^3$)	15.73	8.89	1	10	14	18.9	104.1	2.28	9.43
Şile	Mean	std	Min	0.25	0.5	0.75	Max	Skewness	Kurtosis
Relative Humidity (%)	91.4	10.16	19.2	86.6	94.2	100	100	-1.74	4.36
Air Pressure (hPa)	1003.15	3.56	990.4	1000.8	1003.25	1005.6	1013.2	-0.18	0.1
Air Temperature (°C)	22.69	3.26	11.5	20.5	23.1	25	41.4	-0.25	0.68
Cabin RH (%)	64.72	4.48	48	61.4	64.5	67.8	76.9	0.16	-0.36
Cabin Temperature (°C)	22.7	0.65	18.8	22.3	22.7	23.1	29.1	-0.48	5.73
Wind Speed (m/sn)	3.92	2.02	0.4	2.3	3.7	5.2	12.9	0.71	0.35
Wind Direction (°)	136.36	91.16	6	68.5	96.6	226.5	352.2	0.68	-0.81
Rainfall (mm)	0.01	0.14	0	0	0	0	7	29.23	1153.48
NO_2 ($\mu g/m^3$)	4.26	4.52	0.2	2.5	3	4.3	142.6	8.09	155.25
NO_X ($\mu g/m^3$)	9.08	7.52	1	6.2	7	8.7	165.8	4.68	43.21
O_3 ($\mu g/m^3$)	73.86	23.55	2.9	63	77.9	90	173.6	-0.73	1
PM_{10} ($\mu g/m^3$)	23.65	11.52	1.6	15.9	21.7	28.9	119.1	2.02	8.96

*std = Standard deviation

3.1 XGBoost (Extreme gradient Boosting)

XGBoost (eXtreme Gradient Boosting) is an optimized version of the Gradient Boosting Decision Tree (GBDT) algorithm developed by Chen and Guestrin (2016). This algorithm is widely utilized across various domains due to its ability to enhance computational speed and improve accuracy. The algorithm trains decision trees sequentially, adding a new tree in each iteration to improve model accuracy. The objective function comprises a loss function and a regularization term aimed at reducing model complexity. While the loss function monitors the accuracy of the predictive model, the regularization term acts as a penalty to prevent overfitting. XGBoost facilitates the implementation of cross-validation, a widely used technique to minimize overfitting in each boosting iteration and to determine the error on the validation dataset (Serencam et al. 2022). The objective function is defined as Eq. 1 (Chen and Guestrin 2016):

3.2 K-Nearest neighbor (KNN)

The K-Nearest Neighbor algorithm is a non-parametric model based on the lazy learning method, which does not

Table 3 Descriptive statistics for three AQMS in winter months (December, January and February)

Arnavuktöy	Mean	std	Min	25%	50%	75%	Max	Skewness	Kurtosis
Solar Radiation (W/m^2)	70.67	130.34	0	0	0.1	79.25	670.8	2.1	3.69
Air Pressure (hPa)	1001.62	7.28	980.1	996.7	1001.1	1006.7	1024.4	0.04	-0.24
Air Temperature (°C)	7.68	4.73	-5	4.38	7.7	11	19.6	-0.01	-0.55
Relative Humidity (%)	80.82	13.18	35.7	71.2	81.6	92.4	100	-0.39	-0.59
Wind Speed (m/sn)	4.19	2.34	0.3	2.3	3.8	5.7	13.3	0.74	0.01
Wind Direction (°)	184.01	100.76	0.2	86.2	208.7	258.3	359.9	-0.2	-1.14
Rainfall (mm)	0.05	0.24	0	0	0	0	8	15.59	424.93
NO_2 (µg/m^3)	23.29	18.96	1	9.7	16.9	31.2	125.5	1.54	2.39
NO_X (µg/m^3)	35.86	34.16	2.8	15.3	24.5	43	378.7	3.08	15.57
O_3 (µg/m^3)	37.55	19.71	0.4	23.4	37.55	52.6	97.6	-0.11	-0.69
PM_{10} (µg/m^3)	33.15	23.29	1	17.1	27	41.8	208	1.78	4.61
$PM_{2.5}$ (µg/m^3)	18.98	15.71	1	8.1	14.2	23.9	129.9	1.84	4.25
Bağcılar	Mean	std	Min	25%	50%	75%	Max	Skewness	Kurtosis
Solar Radiation (W/m^2)	50.67	101.43	0	0	0	48.93	658.7	2.42	5.83
Air Pressure (hPa)	1014.5	7.37	992.6	1009.6	1014	1019.6	1034	0.01	-0.25
Air Temperature (°C)	11.24	6.41	-2.6	6.7	10.8	14.6	24.4	0.42	-0.3
Cabin Temperature (°C)	21	4.4	15.1	17.3	19.9	24	29.9	0.49	-0.91
Relative Humidity (%)	70.58	16.95	27.6	61.2	72.6	84.3	100	-0.65	-0.23
Wind Speed (m/sn)	1.6	1.34	0	0.5	1.3	2.4	7	0.84	0
Wind Direction (°)	205.42	109.81	0.1	129.98	177.2	338.2	360	0.05	-1.2
Rainfall (mm)	0.6	2.7	0	0	0	0	59.8	6.7	53.6
NO_2 (µg/m^3)	48.32	29.61	0.6	27.8	42.6	62.8	253.6	1.5	4.16
NO_X (µg/m^3)	111.33	150.24	2.2	38.9	63.1	115.43	1728.5	4.26	24.65
O_3 (µg/m^3)	24.66	18.22	0.4	7.5	22.35	39.1	81.8	0.48	-0.83
PM_{10} (µg/m^3)	41.4	34.46	0.9	21.68	32.8	48.7	400.1	3.12	14.98
Şile	Mean	std	Min	0.25	0.5	0.75	Max	Skewness	Kurtosis
Relative Humidity (%)	85.55	15.3	24.2	74.9	89.45	100	100	-0.91	0
Air Pressure (hPa)	1009.11	7.23	987.1	1004.2	1008.8	1014	1028.2	0.02	-0.23
Air Temperature (°C)	8.46	4.82	-4.6	5.1	8.3	11.4	24.6	0.21	-0.25
Cabin RH (%)	44.84	6.09	30	40.3	44.84	49.1	67.5	0.13	-0.61
Cabin Temperature (°C)	20.35	1.45	12.3	20.2	20.5	21.2	23.6	-1.9	4.95
Wind Speed (m/sn)	5.39	3.27	0.5	2.8	4.7	7.2	19.8	1.05	0.99
Wind Direction (°)	181.59	88.8	4.7	91.08	223.3	245.63	350.4	-0.49	-1.08
Rainfall (mm)	0.06	0.26	0	0	0	0	10	9.58	149.58
NO_2 (µg/m^3)	10.4	9.99	0.3	3.3	7.1	13.8	77.6	1.98	5.19
NO_X (µg/m^3)	15.92	12.29	2.5	7.5	11.9	20	136.9	2.52	11.03
O_3 (µg/m^3)	50.5	18.74	1.5	38	52.8	64.5	99.6	-0.39	-0.41
PM_{10} (µg/m^3)	23.96	12.94	0.2	14.3	22	30.2	122.7	1.25	2.76

*std = Standard deviation

make any prior assumptions about the data and performs the learning process at the prediction stage, not at the model creation stage (Yao and Ruzzo 2006). Therefore, this model is important for air pollution prediction (Choubin et al. 2020). The basic idea of this method is to find k historical series that are most similar to the predicted one. The steps involved in predicting univariate time series with kNN are vector generation and similarity estimation and prediction. KNN algorithms have been widely researched (Goh and Ubeynarayana 2017; Soh et al. 2018; Wang et al. 2021; Li et al. 2023; Kigo et al. 2023). However, their results under various geographical and meteorological conditions are still questionable. The objective function is defined as Eq. 2 (Cover and Hart 1967):

3.3 Extra trees (Extremely randomized trees)

Decision trees are used to predict the value of a discrete dependent variable, which belongs to a finite set of values (referred to as “classes”). The independent variables, often termed as “features,” can be either continuous or discrete. These features are utilized to predict continuous values in regression problems or discrete classes in classification problems. Decision trees, also known as top-down inductive decision tree algorithms, employ a divide-and-conquer approach (Quinlan 1986). The Extra Trees Regressor primarily follows the classic top-down process and utilizes the entire training dataset for constructing the trees (Mannodi-Kanakkithodi et al. 2016). Prediction is obtained by averaging across trees in Eq. 3:

3.4 Random forest (RF)

The RF model based on machine learning creates a certain number of decision trees. Then, it creates random forests by classifying these decision trees according to certain criteria. This RF model can be up to thousands due to the existence of multi-layered processes. This model can be used to simulate multiple non-linear relationships (Biau and Scornet 2016).

3.5 Adaptive boosting (AdaBoost)

AdaBoost combines multiple weak learners by updating the weights of the examples during the training process. These weak learners are regression trees (Barrow and Crone 2016). At each iteration, the algorithm rearranges the weights of the examples according to their prediction performance; the weights of the examples with high prediction errors are increased, while the weights of the examples with more accurate predictions are decreased. As each new tree is added, the effect of the previous model decreases and the strongest tree is included in the model (Eq. 2). Through this iterative process, the performance of the model improves over time. However, the initial base model may classify some examples correctly and others incorrectly. AdaBoost is a weak learner boosting method that improves this situation with continuous training, increasing the classification performance (Schapire 1990).

3.6 Support vector regression (SVR)

Support Vector Regression (SVR) is a method that applies the concept of SVR to regression problems, aiming to find a function that deviates from the actual target values by a margin no greater than a specified threshold while maximizing generalization (Drucker et al. 1997). The SVR method demonstrates strong performance in time series and regression tasks (Müller et al. 1997). The SVR model relies on a subset of the training data by ignoring all data points that fall within a threshold (ε) close to the model's prediction (Suárez Sánchez et al. 2011). One of the strengths of SVR lies in its ability to map nonlinear data into a high-dimensional space using a linear function (Bai et al. 2018). The optimization problem is defined in Eq. 5:

3.7 The Multi-Layer perceptron (MLP)

The Multi-Layer Perceptron (MLP) is a type of NN characterized by a layered structure where neurons establish full connections between layers. Each neuron in the intermediate layer's forms complete connections with all neurons in the preceding layer. At the neuron level, a nonlinear transformation is applied, and the resulting outputs are transmitted to the next layer. The training of an MLP is performed using the backpropagation algorithm, which aims to minimize a defined loss function (Özdoğan, 2020). This transformation is expressed in Eq. 2.

3.8 Shapley additive explanation method

After determining the best model, SHapley Additive Explanations (SHAP) method was used to visualize the formatting and variability of the expansion of the model. SHAP is used to determine how much an individual predictor contributes to the outcome of the predictive model (Lundberg and Lee 2017; Başakın et al. 2024). This approach is based on Shapley (Shapley 1953), which is based on the game theory. SHAP summary graphs show the overall order of contribution of the features to the model and the direction of this contribution (positive/negative) for an observation. Hence, the SHAP algorithm enhances model interpretability by identifying and quantifying the contribution of meteorological variables to the variation in air pollutant concentrations (Friedman 2001; Goldstein et al. 2015). The larger the absolute SHAP value of the features, the more pronounced their effect on the model prediction.

3.9 Performance evaluation

Many approaches are used to evaluate the results of ML methods. To ensure consistency across all performance metrics, the observed pollutant concentrations are denoted as O_i, while the model-predicted concentrations are denoted as P_i. The overbar symbol (e.g., $\overline{O}$, $\overline{P}$) represents the arithmetic mean of the corresponding variable. The first of these is the Root Mean Square Error (RMSE), which is the standard deviation of the prediction errors. RMSE is a measure of how far these errors are from the data points on the regression line (Chai and Draxler 2014) (Eq. 1). Index of Agreement (IoA): A fit index (d) was proposed as a standard measure of the degree of model prediction error ranging from −1 to 1. If the closer it is to 1, the higher the probability of model success (Willmott 1981) (Eq. 2). Mean absolute error (MAE) is primarily used to estimate the mean deviation in the model (Eq. 3). Also, the Nash–Sutcliffe Efficiency (NSE) index (Nash and Sutcliffe 1970) and the determination coefficient (R^2) were used (Eqs. 4 and 5). R^2

is a widely used performance metric across various studies; it was examined to evaluate the accuracy of the prediction outcomes. A value of 0 indicates poor performance, while 1 signifies excellent performance. Finally, the Kling-Gupta efficiency (KGE) (Gupta et al. 2009) was used (Eq. 6). After the prediction results, RMSE, MAE, BIAS, NSE, KGE, and R^2 tests were applied for the predicted PM_{10}, $PM_{2.5}$, and O_3 concentrations to determine the performance.

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(P_i - O_i)^2}{n}} \quad (1)$$

$$d_r = \begin{cases} 1 - \frac{\sum_{i=1}^{n} |P_i - O_i|}{c\sum_{i=1}^{n} |O_i - \bar{O}|},\ when \\ \sum_{i=1}^{n} |P_i - O_i| \leq c\sum_{i=1}^{n} |O_i - \bar{O}| \\ \frac{c\sum_{i=1}^{n} |O_i - \bar{O}|}{\sum_{i=1}^{n} |P_i - O_i|} - 1,\ when \\ \sum_{i=1}^{n} |P_i - O_i| > c\sum_{i=1}^{n} |O_i - \bar{O}| \end{cases} \quad (2)$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N} |P_i - O_i| \quad (3)$$

$$NSE = 1 - \frac{\sum_{i=1}^{n} (P_i - O_i)^2}{\sum_{i=1}^{n} (O_i - \bar{O})^2} \quad (4)$$

$$R^2 = \frac{\left[\sum_{i=1}^{n} [(O_i - \bar{O})(P_i - \bar{P})]\right]^2}{\sum_{i=1}^{n} [O_i - \bar{O}]^2 \sum_{i=1}^{n} [P_i - \bar{P}]^2} \quad (5)$$

$$KGE = 1 - \sqrt{(R-1)^2 + (\propto -1)^2 + (\beta - 1)^2}, \\ where\ R = corr(O_i, P_i),\ \propto = \frac{\sigma_P}{\sigma_O},\ \beta = \frac{\bar{P}}{\bar{O}} \quad (6)$$

4 Results and discussions

Figures 2, 3 and 4 illustrate the distribution of residuals across training and test sets for different model-season configurations. The violin plots were specifically chosen to depict not only the central tendency, as in boxplots, but also the spread and density of residuals. Narrow and centered distributions correspond to stable model behavior with limited prediction error, while broader shapes reflect higher uncertainty or systematic bias. These distributional features serve as a diagnostic tool to identify models with stronger generalization ability under varying meteorological and emission regimes. Figure 2 compares the residual distributions of PM_{10} estimates for Bağcılar station in (a) winter and (b) summer. In both seasons, the training residuals are narrow and centered around zero, indicating good model fit. However, the test residuals in winter are more compact and less dispersed compared to summer, indicating better generalization under stable winter meteorological conditions. The summer panel exhibits wider and more variable test residuals, probably due to complex pollutant dynamics and weaker seasonal patterns. These results highlight the seasonal dependence of model reliability for urban PM_{10} estimation. Similarly, when seasonal separation is applied in PM_{10} estimation, it is found that the residual distribution changes fluctuation and residual heterogeneity become evident, especially in summer (Gregório et al. 2022). Also, Kalantari et al. (2025) stated that the RF model offers higher R^2 and lower residual variance in the summer season.

The graphical evidence aligns with the quantitative metrics reported in Tables 4 and 5. For instance, the compact residual distributions in the winter season are consistent with the higher R^2 and lower RMSE values in Table 4, whereas the broader and more skewed distributions in summer reflect the weaker predictive skill reported in Table 5. Thus, Figs. 2, 3 and 4 provide a visual complement to the statistical summaries, facilitating a more intuitive interpretation of model robustness across time and space.

Figure 3 illustrates the distribution of residuals from $PM_{2.5}$ predictions for Bağcılar station in (a) winter and (b) summer. In winter, test residuals are more tightly clustered around zero, indicating stronger model generalization and less variability. In contrast, the summer panel shows a broader spread in residuals, reflecting increased uncertainty likely due to more complex emission dynamics and meteorological variability. Despite this, the overall bias remains low, as most residuals are symmetrically distributed. These results reinforce the seasonal dependency of $PM_{2.5}$ predictability and highlight the enhanced stability of wintertime estimations. Likewise, Aman et al. (2024) showed in their seasonal ML analysis in Bangkok that the residual distribution of $PM_{2.5}$ is narrower and more stable in winter, while the distribution widens in summer due to meteorological and emission variability.

Figure 4 shows the residual distributions of O_3 predictions for (a) Arnavutköy and (b) Bağcılar stations during the summer season. While both stations exhibit symmetric residual distributions centered near zero, notable differences emerge in dispersion. In Arnavutköy, residuals are more widely spread, especially for test data, indicating higher uncertainty likely due to complex rural influences such as biogenic emissions, long-range transport, and weaker local emission signals. In contrast, Bağcılar demonstrates narrower residual ranges, suggesting more stable and learnable

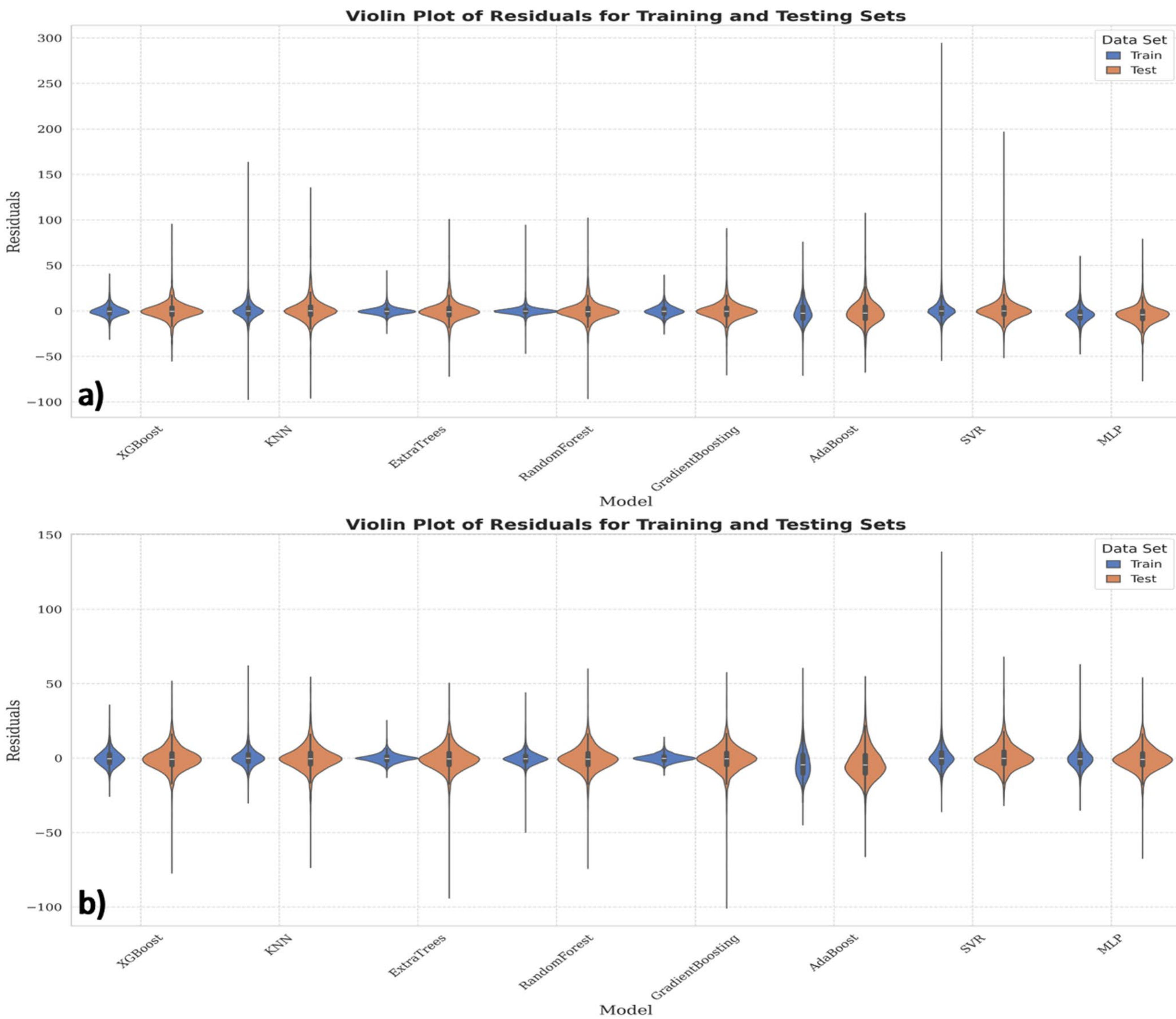

Fig. 2 Violin Plot of Training and Testing sets for PM_{10} (**a**) Winter (DJF), (**b**) Summer (JJA) in Bağcılar (urban)

patterns, consistent with its urban characteristics and dominant anthropogenic sources. These findings highlight the impact of station typology on model behavior, particularly for secondary pollutants like ozone. Although O_3 predictions were generally accurate, increased dispersion in rural stations such as Arnavutköy may reflect the influence of unaccounted episodic sources or regional transport, as similarly discussed by Calatayud et al. (2023) for PM-related uncertainties.

To prevent temporal leakage, all evaluations were conducted using fixed seasonal test splits (two summers and two winters), rather than random folds. Thus, each model was trained and validated on one seasonal subset and then evaluated on a distinct held-out test season. Because of this deterministic partitioning, we did not compute fold-based confidence intervals. Instead, robustness was assessed through consistent performance across three stations and two contrasting meteorological regimes. As an additional baseline, a multiple linear regression model was also tested, which produced systematically lower R^2 values and higher RMSE compared to the ML algorithms, confirming that the reported improvements are meaningful. Finally, while Table 4 highlights the best-performing model for each pollutant-site combination, performance gaps among the leading algorithms were generally modest, indicating that model superiority is supported by multi-metric consistency rather than by isolated single-metric gains. The estimation results for PM_{10}, $PM_{2.5}$, and O_3 concentration in winter months were also made and the best method was determined as a result of diligent comparison. Table 4 shows the train and test results for the three districts in the winter period.

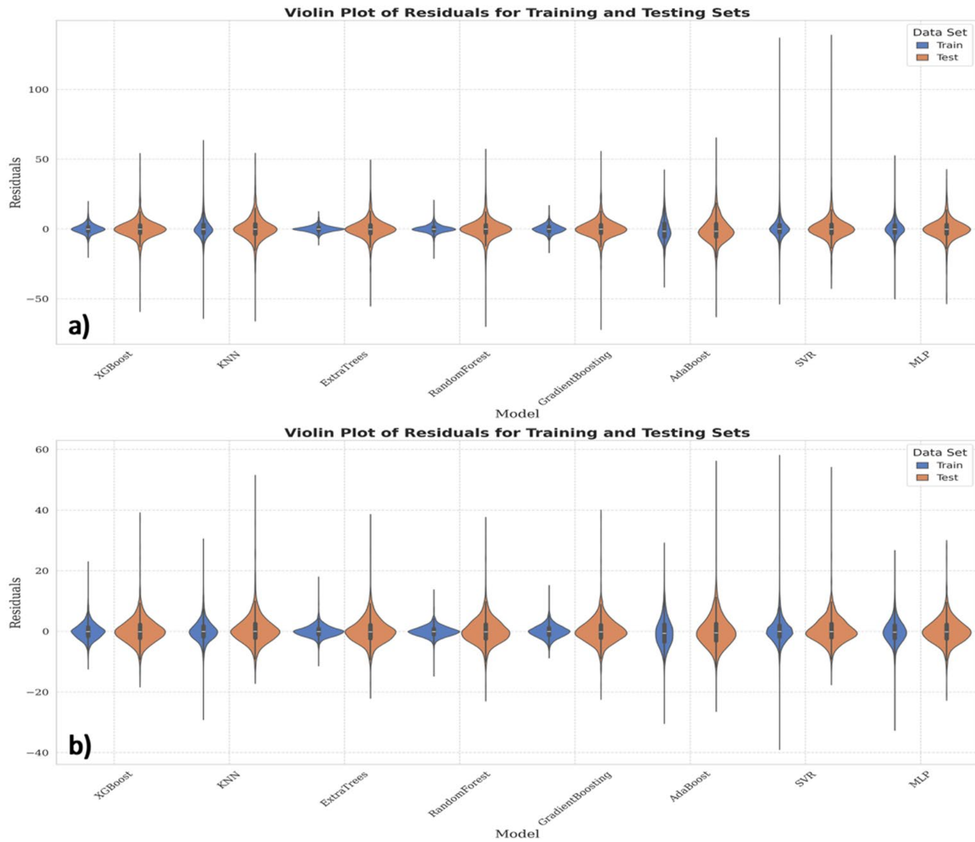

Fig. 3 Violin Plot of Training and Testing sets for $PM_{2.5}$ Winter (DJF), **b** Summer (JJA) in Bağcılar (urban)

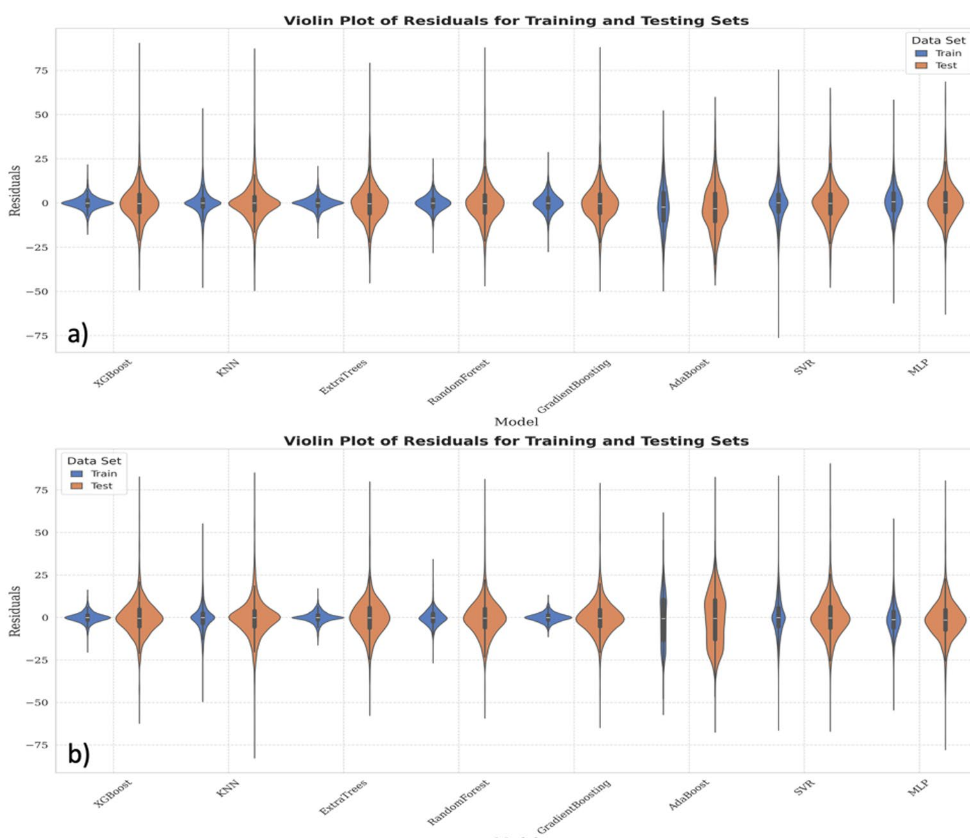

Fig. 4 Violin plots of residuals for O_3 predictions at **a** Arnavutköy (rural) and **b** Bağcılar (urban) stations during the summer (JJA)

Table 4 Performance metrics of Best-Performing models for PM_{10}, $PM_{2.5}$, and O_3 during the winter period (DJF)

	Station	Model	Category	MAE	RMSE	R^2	Willmott d1	KGE	NSE
PM_{10}	Arnavutköy	XGBoost	Training	3.27	4.54	0.96	0.9	0.95	0.96
			Testing	5.83	8.6	0.86	0.82	0.9	0.86
	Bağcılar	XGBoost	Training	4.21	5.7	0.97	0.9	0.97	0.97
			Testing	6.36	9.93	0.93	0.86	0.95	0.93
	Şile	ExtraTrees	Training	5.48	8.49	0.92	0.7	0.86	0.92
			Testing	6.63	10.37	0.65	0.6	0.6	0.65
$PM_{2.5}$	Arnavutköy	XGBoost	Training	2.41	3.23	0.96	0.89	0.95	0.96
			Testing	4.48	6.45	0.83	0.8	0.88	0.83
	Bağcılar	XGBoost	Training	2.40	3.22	0.98	0.92	0.98	0.98
			Testing	4.47	6.67	0.91	0.85	0.94	0.91
O_3	Şile	KNN	Training	2.74	4.47	0.94	0.91	0.95	0.94
			Testing	4.54	7.18	0.86	0.85	0.91	0.86
	Arnavutköy	KNN	Training	3.70	5.91	0.91	0.88	0.93	0.91
			Testing	5.75	8.97	0.80	0.82	0.87	0.80
	Bağcılar	XGBoost	Training	1.41	2	0.98	0.95	0.97	0.98
			Testing	4.10	5.99	0.89	0.86	0.91	0.89

Table 5 Performance metrics of Best-Performing models for PM_{10}, $PM_{2.5}$, and O_3 during the summer period (JJA)

	Station	Model	Category	MAE	RMSE	R^2	Willmott d1	KGE	NSE
PM_{10}	Arnavutköy	ExtraTrees	Training	4.04	9.44	0.78	0.87	0.88	0.95
			Testing	11.11	18.83	0.68	0.62	0.55	0.51
	Bağcılar	MLP	Training	4.85	6.73	0.86	0.79	0.88	0.86
			Testing	5.66	7.76	0.8	0.76	0.88	0.8
	Şile	KNN	Training	3.49	5.57	0.77	0.78	0.82	0.77
			Testing	5.18	7.54	0.63	0.65	0.69	0.56
$PM_{2.5}$	Arnavutköy	RandomForest	Training	1.5	2.47	0.89	0.84	0.85	0.93
			Testing	3.34	4.9	0.67	0.63	0.68	0.6
	Bağcılar	MLP	Training	2.84	3.82	0.81	0.75	0.86	0.81
			Testing	3.05	4.15	0.8	0.75	0.86	0.8
O_3	Bağcılar	XGBoost	Training	2.2	3.03	0.99	0.95	0.97	0.99
			Testing	7.08	9.93	0.85	0.82	0.88	0.85
	Şile	XGBoost	Training	2.92	4.07	0.97	0.92	0.95	0.97
			Testing	7.50	10.35	0.81	0.78	0.87	0.81
	Arnavutköy	KNN	Training	4.39	6.74	0.92	0.91	0.94	0.92
			Testing	6.61	10.38	0.78	0.8	0.84	0.78

The model performances in winter were significantly strong across all pollutants and stations. For PM_{10}, the XGBoost model showed excellent generalization ability with test R^2 values of 0.86 and 0.93 in both Arnavutköy and Bağcılar, respectively. Similarly, Extra Trees achieved a high test R^2 of 0.89 in Şile, indicating reliable prediction accuracy. $PM_{2.5}$, the test R^2 values exceeded 0.80 in all locations. Likewise, PM_{10} and $PM_{2.5}$ predictability in winter months is stronger than in summer months. The XGBoost and Extra Trees models showed robust performance, especially in Bağcılar and Arnavutköy. These models showed minimal overfitting and strong compatibility between training and testing metrics. The models also showed competitive results for O_3. The XGBoost model had better results than other models used in India (Van et al. 2023). XGBoost in Arnavutköy and Bağcılar yielded test R^2 scores of 0.80 and 0.88, respectively. Although the test R^2 was slightly lower (0.71), the Extra Trees model still captured O_3 variability reasonably well in Şile. Also, The XGBoost model shows high accuracy and strong generalization ability in predicting both $PM_{2.5}$ and O_3 concentrations in China (Wang et al. 2023); such that the R^2 value in the test data for $PM_{2.5}$ is 0.912 and for O_3 is 0.854, which shows that the model account for nonlinear relationships. Ensemble-based models such as XGBoost and Extra Trees provided consistent and accurate prediction throughout the winter, with $PM_{2.5}$ and O_3 generalizing slightly better than PM_{10} This is mainly because the coarse fraction of PM_{10} (PM_{10} - $PM_{2.5}$) originates from localized and episodic sources such as dust resuspension, construction, and agricultural activities, which create strong spatial variability, unlike the more chemically homogeneous $PM_{2.5}$ fraction (Li et al. 2013). Table 5 shows the train and test results for the summer months. In the present work, only the best-performing models for each pollutant and station are presented (Tables 4 and 5). The complete performance results of all seven ML models are provided as supplementary material (Table S3 and S4 for the summer period and Table S5 and S6 for the winter period).

During the summer period, model performance varied across pollutants and stations. PM_{10} predictions showed relatively lower accuracy, especially in Arnavutköy, where the Extra Trees model suffered low predictive performance (train $R^2 = 0.63$; test $R^2 = 0.61$). This reduced performance is likely linked to the greater spatial variability of coarse particles (PM_{10} and $PM_{2.5}$), which are strongly influenced by local resuspension, dust transport, and mechanical processes (e.g., traffic-induced resuspension, agricultural activities). Such episodic and site-specific sources make PM_{10} more difficult to predict compared to the more chemically consistent $PM_{2.5}$ fraction (Wagstrom and Pandis 2011). In contrast, the MLP model in Bağcılar gave consistent results (train $R^2 = 0.86$; test $R^2 = 0.80$), while the KNN in Şile provided moderate test performance ($R^2 = 0.63$). For $PM_{2.5}$, both Arnavutköy and Bağcılar showed better generalization. Random forest in Arnavutköy showed high accuracy (test $R^2 = 0.89$), whereas the MLP model in Bağcılar showed robust performance (test $R^2 = 0.80$). In a study conducted in Aksaray, Türkiye, the Extra Trees model was found to be highly successful in predicting PM_{10} (Koçak 2025). O_3 predictions were remarkably successful, with XGBoost in Bağcılar achieving the best overall test R^2 (0.85). KNN performed well in Arnavutköy ($R^2 = 0.74$) and reasonably well in Şile ($R^2 = 0.81$). XGBoost and RF gave the best results in urban areas ($R^2\% \approx 0.83$–0.88), while in rural regions O_3 modeling accuracy remained at $R^2 \approx 0.70$–0.75 (Pan et al. 2023). Also, it shows that ozone (O_3) concentrations in China are strongly related to seasonal effects, which is similar to our study with increased prediction success in summer months (Dai et al. 2023a, b). In summary, ensemble models such as Extra Trees and XGBoost have proven particularly effective for O_3 and $PM_{2.5}$, while PM_{10} predictions remain more challenging due to spatial variability and risks of overfitting.

Tables 4 and 5 present the best-performing site-pollutant combinations for each season rather than listing all stations, in order to emphasize the most representative and robust results. According to Tables 4 and 5, model performances varied across seasons and station types. In winter months, due to more stable atmospheric conditions, the prediction accuracy for PM_{10} and $PM_{2.5}$ increased. In Bağcılar, an urban station, heavy traffic and regular emissions from construction enabled the models to produce more successful results. In Arnavutköy, a rural location, transport and biogenic effects increased the uncertainty reflected in the model in the summer months. Şile, an urban background, generally showed moderate success. Model accuracy was higher at urban and traffic-impacted sites (e.g., Bağcılar) than at rural sites (e.g., Şile). This is because urban concentrations are primarily governed by strong and local emission sources, which provide a more stable relationship between predictors and pollutants. In contrast, rural sites are more influenced by background meteorology and long-range dust transport, leading to greater variability and reduced model predictability. Likewise, according to literature employing different ML methods, it was found that air quality prediction in urban areas was more successful than in rural areas (Jayaraman 2025). Also, a comparative evaluation of SVR, RF, and XGBoost models for $PM_{2.5}$ distribution prediction under different seasonal conditions in Malaysia found that the RF model outperformed XGBoost in the overall accuracy (Zaman et al. 2024). This situation supports that the emission source density and meteorological conditions directly affect model success. Therefore, station-specific modeling strategies should be developed by taking locational differences into account. Overall, the training performance was higher than the test performance, which is expected due to model fitting on the training data. The observed gaps between training and testing sets are modest for most cases, indicating good generalization. Larger discrepancies (e.g., O_3 in summer at rural sites) reflect higher variability and non-stationarity in pollutant-meteorology relationships, making prediction more challenging under unseen conditions.

Figure 5 shows the scatter plots of PM_{10} predictions using XGBoost and Extra Trees models for three different stations in winter. The XGBoost model (Fig. 5) demonstrated excellent generalization with tightly clustered test predictions along the ideal line, indicating strong agreement between training and testing performances. In contrast, the Extra Trees model (Fig. 5c) exhibited lower generalization ability, with testing results showing wider dispersion from the 1:1 ideal line, indicating reduced reliability compared to the XGBoost, particularly under varying seasonal conditions. It was also emphasized that the spatial accuracy remained low in some regions and the model success was affected by seasonal and regional factors as illustrated by Park et al. (2021). These results highlight the sensitivity of PM_{10} prediction to both model complexity and station-specific conditions, particularly in rural areas during summer.

Figures 6 shows $PM_{2.5}$ prediction performance during summer using MLP and RF models for Bağcılar and Arnavutköy, respectively. The MLP model achieved strong generalization in the urban setting, with both training and testing predictions closely aligned with the ideal line. Singh and Suthar (2025) employed different ML algorithms in Jaipur, and as a result, deep learning-based CNN and ANN models

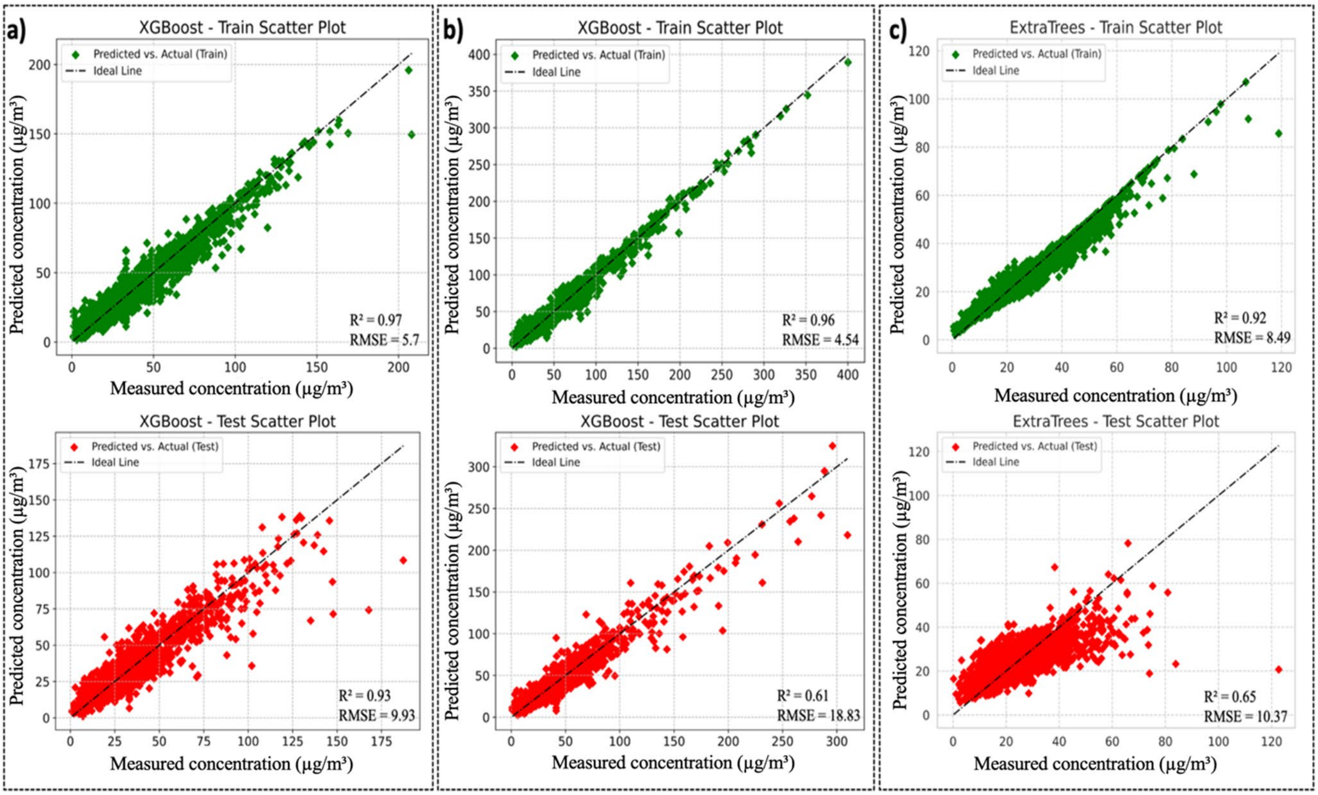

Fig. 5 Scatter plots of actual vs. predicted PM_{10} concentrations using the best-performing models at three stations in winter: **a** Bağcılar (urban), **b** Arnavutköy (rural), and (**c**) Şile (urban background)

provided high accuracy compared to their counterparts. Also, in $PM_{2.5}$ prediction in China, Bayesian neural network achieved MLP ($R^2 \approx 0.70$) in urban environments. In contrast, the Extra Trees model (Fig. 7) demonstrated signs of overfitting in Arnavutköy, while training results were nearly perfect and testing predictions were significantly scattered, indicating reduced robustness in the rural areas with respect to the corresponding ML algorithm. While high accuracy is achieved in the autumn months in the Middle East sample, PM estimates in the winter months in our study provide more reliable results. This shows that meteorological stability directly contributes to model generalization (Talepour et al. 2024). Moreover, NN-based models showed superior performance in $PM_{2.5}$ prediction ($R = 0.949$) in Delhi, while the accuracy increased in urban areas (Masood and Ahmad 2023). Similarly, the MLP model provided high generalization capacity during summer months in Bağcılar. These findings highlight the influence of station characteristics on model stability and performance as further supported by the current analysis.

Figures 7 presents the summer O_3 prediction results using the XGBoost and KNN model in Bağcılar, Arnavutköy and Şile. In Bağcılar, the model achieved excellent predictive performance, with both training and test results showing a strong linear trend along the ideal line and minimal dispersion. RFNN-GOA model, ozone prediction performance differed between urban and rural stations by Braik et al. (2024). Similarly, integrated CTM-ML approaches have demonstrated excellent performance in O_3 prediction across Italy, where Random Forest improved spatial resolution and reduced biases compared to CTM alone (Silibello et al. 2021). In contrast, the model's predictions in Arnavutköy were slightly more scattered in the test set, indicating moderate generalization in rural areas where background variability may be higher. Similar results were reported for the SoCAB, where ML models predicted O_3 reliably in urban areas but performed poorly at rural edge sites due to higher variability (Do et al. 2024). Also, In Mexico City, CNN-based models achieved very high agreement (d ≈ 0.91–0.93) in predicting daily O_3, confirming the suitability of ML for reliable ozone prediction in complex urban environments (Ahmad et al. 2025). In Şile, although the XGBoost model maintained relatively high accuracy, the test results revealed a slight underestimation at higher concentration levels, suggesting that complex coastal dynamics and local meteorological influences may reduce model robustness in rural coastal environments.

As shown in Fig. 8, the XGBoost model achieved robust predictive performance for $PM_{2.5}$ at Bağcılar, with high R^2 values and a close alignment along the 1:1 ideal line. In

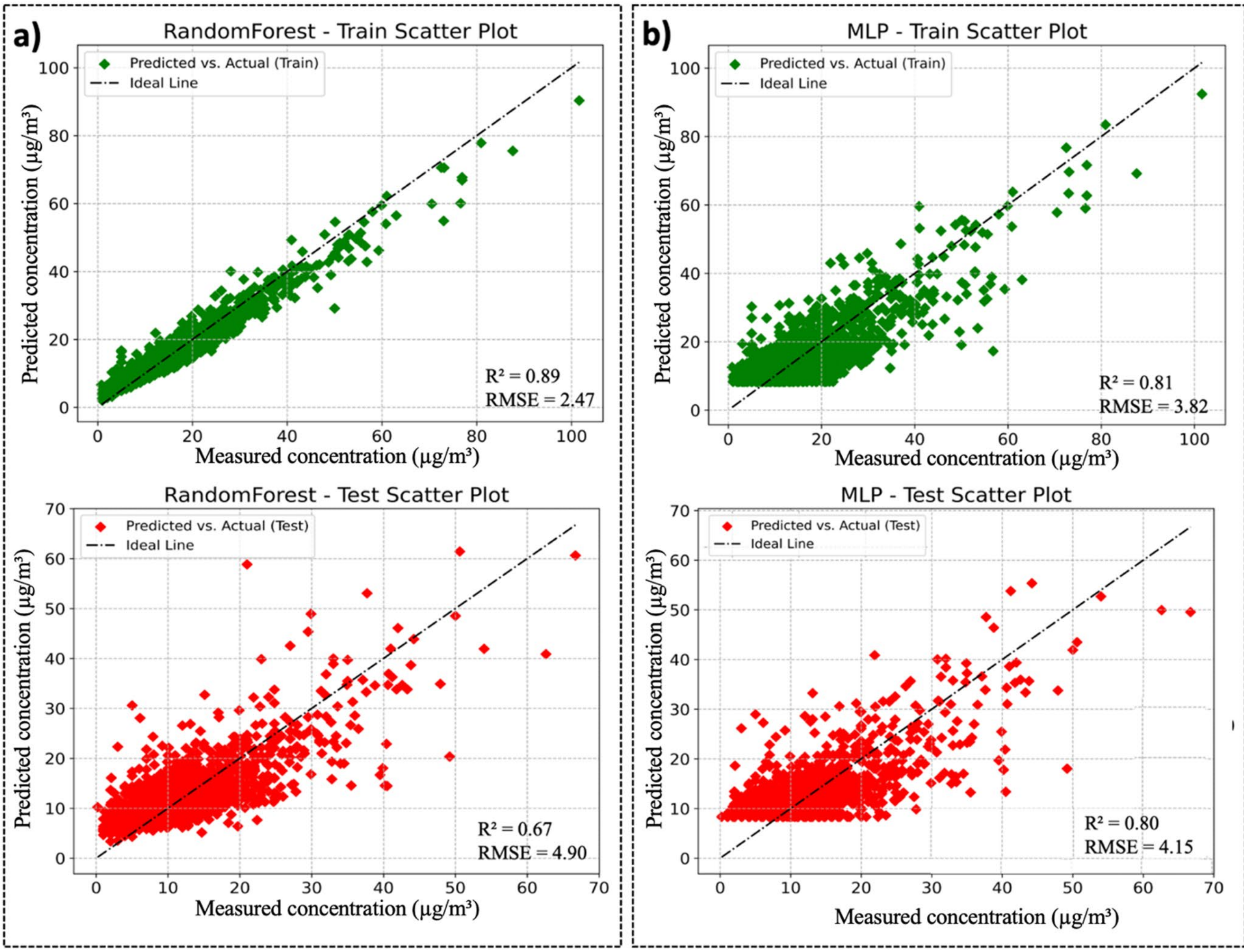

Fig. 6 Scatter plots of actual vs. predicted $PM_{2.5}$ concentrations using the best-performing models at two stations in summer: **a** Bağcılar (urban) and **b** Arnavutköy (rural)

contrast, the results at Arnavutköy revealed a slightly wider scatter in the test set, suggesting that rural background variability reduced the model's generalization ability. According to Fig. 9, PM_{10} prediction performance varied across the three stations. The strongest agreement between observed and predicted values was achieved at Bağcılar, where both training and test results showed high consistency. In Arnavutköy, moderate deviations appeared in the test set, reflecting local background effects. At Şile, test predictions underestimated higher concentration levels, likely due to coastal meteorological influences that limited model accuracy. Hence, in the current research, we only reported the results of the best-performing models for each pollutant and station to maintain clarity and focus. However, for completeness, the outputs of all the other models that were tested are also provided in the Supplementary Material (Figs. S1-S15). These additional plots allow for the comparison of the relative performance of different algorithms across pollutants (PM_{10}, $PM_{2.5}$, and O_3), stations (Bağcılar, Arnavutköy, and Şile), and seasons (winter and summer).

In this study, after determining the best model, a detailed analysis was performed for three different stations and pollutant types in order to increase model interpretability with the SHAP method. SHAP analyses were performed using the XGBoost models, which gave the best results in estimating $PM_{2.5}$ and PM_{10} at Bağcılar station in winter, and the RF model was used in estimating $PM_{2.5}$ at Arnavutköy station in summer. While PM_{10} and $PM_{2.5}$ are inherently correlated, both were retained to capture co-pollutant interactions, and SHAP results were interpreted with this dependency in mind. These cases were chosen to highlight urban-rural contrasts and seasonal variability in variable contributions, providing insight into both spatial and temporal dynamics. These examples were selected to reveal which meteorological variables the model evaluates in areas with both urban and rural characteristics. Thus, spatial and seasonal differences in variable contributions were examined in depth.

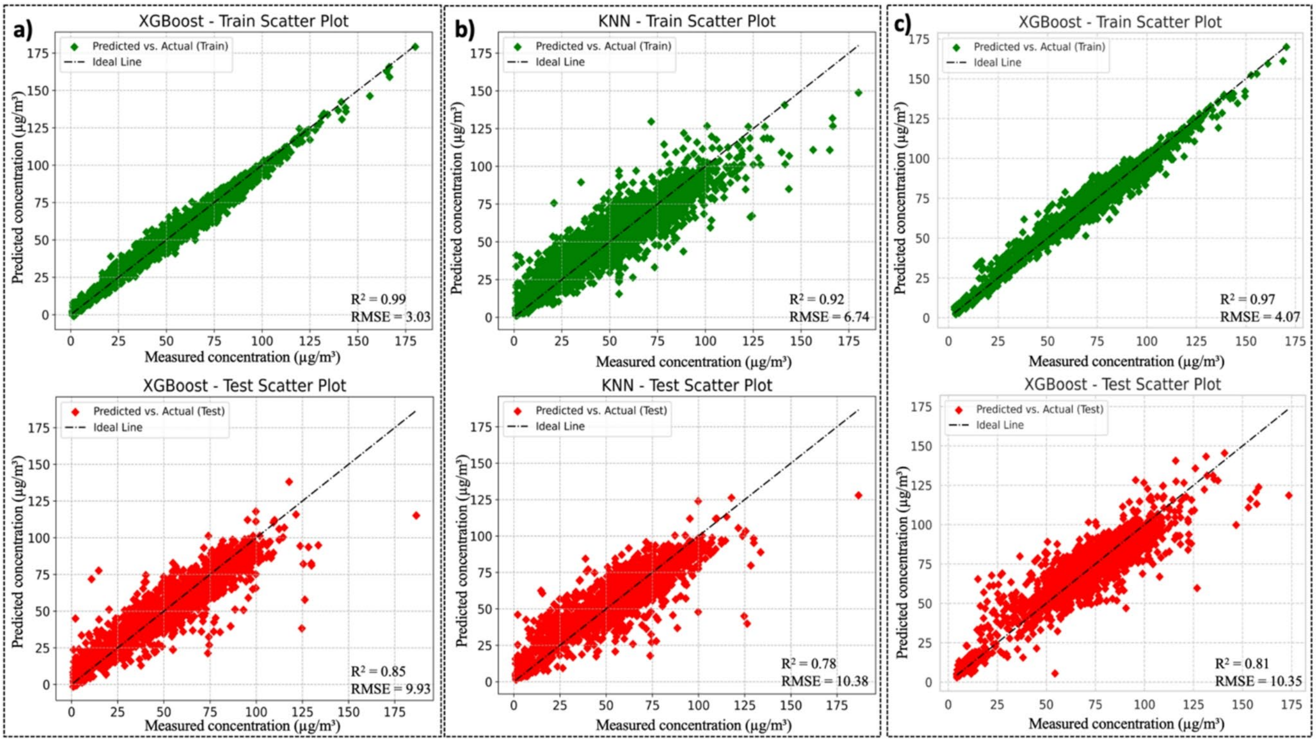

Fig. 7 Scatter plots of actual vs. predicted O_3 concentrations using the best-performing models at three stations in summer: **a** Bağcılar (urban), **b** Arnavutköy (rural), and **c** Şile (urban background)

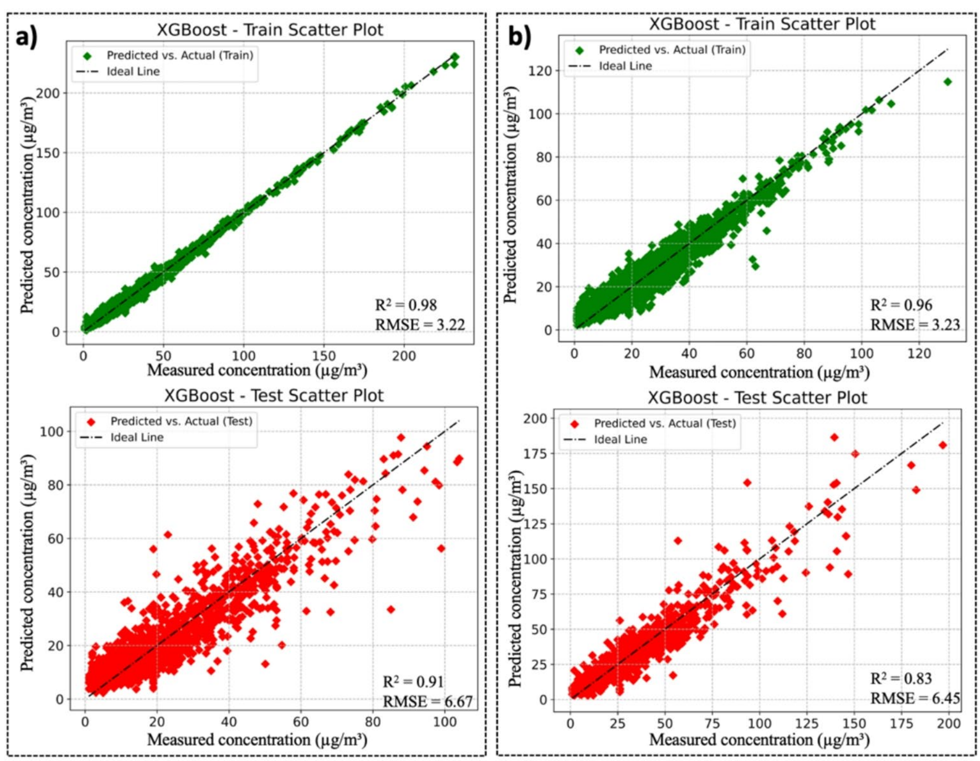

Fig. 8 Scatter plots of actual vs. predicted $PM_{2.5}$ concentrations using the best-performing models at two stations in winter: **a** Bağcılar (urban) and **b** Arnavutköy (rural)

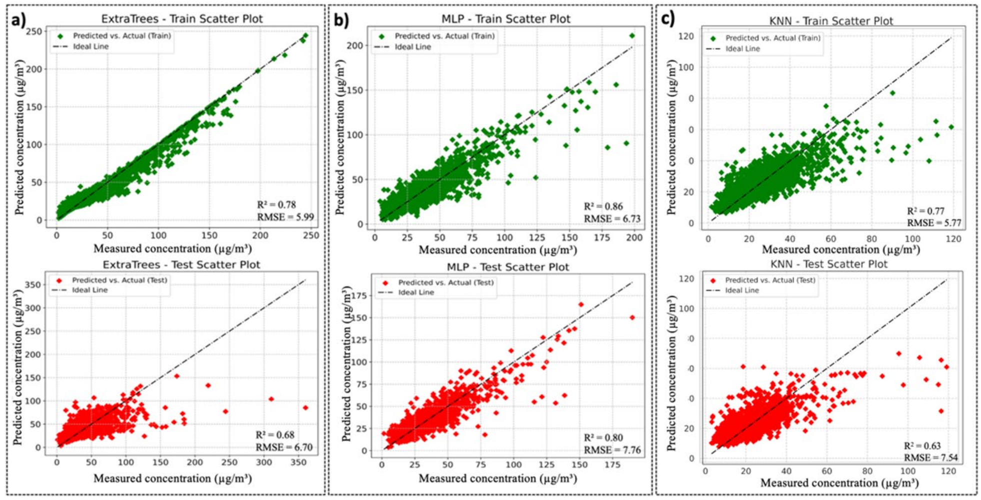

Fig. 9 Scatter plots of actual vs. predicted PM_{10} concentrations using the best-performing models at three stations in summer: **a** Bağcılar (urban), **b** Arnavutköy (rural), and (**c**) Şile (urban background)

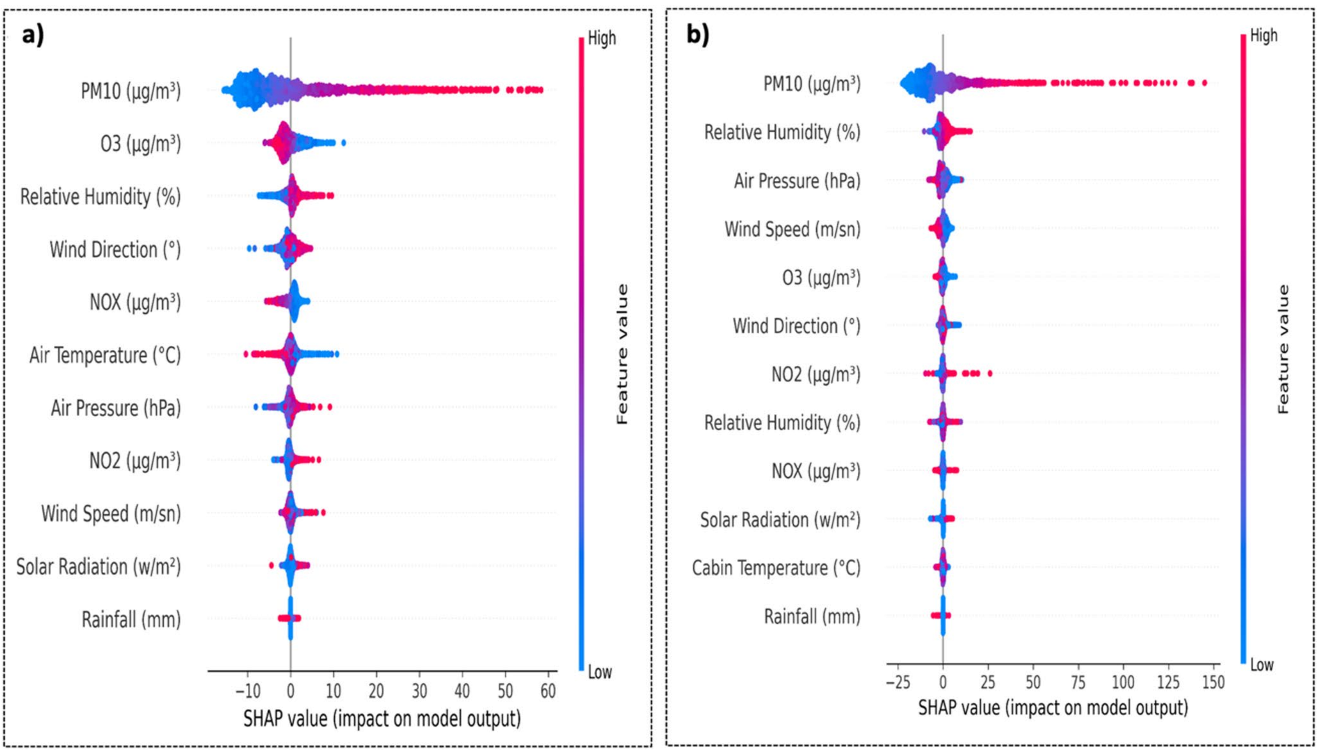

Fig. 10 Variable importance ranking obtained with SHAP for the prediction of $PM_{2.5}$ concentrations during the winter period at **a** Arnavutköy (rural), **b** Bağcılar (urban) AQMS

According to the SHAP analysis results for the Bağcılar station in winter, in Fig. 10, it is seen that PM_{10} is by far the most effective variable in estimating $PM_{2.5}$ concentrations. This dominance is not only statistically expected but also chemically meaningful, since $PM_{2.5}$ is physically contained within PM_{10}, and coarse particle variability (e.g., resuspension, dust events) introduces additional complexity. This situation reflects the strong correlation between particulate matter types. It is noteworthy that meteorological variables such as relative humidity (%), air temperature (°C) and wind speed (m/s) are effective after PM_{10}. This finding is consistent with previous studies, which have shown that meteorological factors such as dew point and temperature strongly influence $PM_{2.5}$ levels in other regions, for example in China (Zhang et al. 2023a, b). In our case, while dew point was not included as a predictor, the role of humidity and temperature aligns with the physical mechanisms identified in that study. Among the meteorological factors included into SHAP analysis, wind speed played a crucial role in determining $PM_{2.5}$ concentration (Liu et al. 2025). It was found that NO_2 and NO_x also make significant contributions to the model. SHAP contributions are positive as they increase $PM_{2.5}$ levels, especially in cases where NO_2 values are high.

The study conducted by García and Aznarte (2020) and utilized the SHAP technique estimated NO_2 concentrations in Madrid, Spain, and found that wind speed had the greatest impact on the estimates. This study also adopted the SHAP analysis to the XGBoost model developed for the winter $PM_{2.5}$ estimation for Arnavutköy and Bağcılar AQMS (Fig. 10). The contribution of variables like wind direction and pressure was found as relatively limited. Additionally, this analysis shows that the dominant factors determining $PM_{2.5}$ levels in Bağcılar are both particle concentration and meteorological conditions. A recent study in Bangkok used SHAP analysis together with meteorological normalization to separate emission and meteorology-driven contributions to $PM_{2.5}$, showing that while meteorological factors significantly modulate pollutant levels, emissions remain the dominant driver under many conditions (Aman et al. 2025).

SHAP analysis was further applied to the XGBoost and extra tree model developed for the winter PM_{10} estimation for Arnavutköy, Bağcılar and Şile AQMS (Fig. 11). This reflects the strong correlation between $PM_{2.5}$ and PM_{10}. However, air temperature can also affect PM_{10} accumulation, especially under inversion conditions. The positive SHAP effects of relative humidity indicate that particle accumulation increases in humid weather. While the effects of variables such as wind speed and direction are more limited in the model, the contributions of gaseous pollutants such as NO_x and NO_2 are generally low but positive. This analysis confirms that particle pollution in urban areas is closely related to meteorological conditions, especially fine particles. In a study conducted in İstanbul, it was found that the minimum number of variables to estimate PM_{10} concentration was 6 (Birinci et al. 2025). Therefore, it is essential to use the variables together for PM_{10} estimation. A limitation of this study is that SHAP results were not validated with ablation tests, but the observed importance patterns were consistent with known atmospheric mechanisms, which supports their interpretability.

SHAP analysis of $PM_{2.5}$ estimation made with RF and MLP model for Arnavutköy and Bağcılar AQMS in summer season clearly reveals the dominant effect of PM_{10} in the estimations (Fig. 12). After PM_{10}, pollutants such as wind speed and direction and NO_2 are at the top of the ranking, and it can be understood that these variables affect $PM_{2.5}$ concentrations especially due to transport. Meteorological variables such as relative humidity, wind speed and temperature are effective on atmospheric dilution or accumulation of $PM_{2.5}$ and have made a moderate contribution. In addition, the effect of variables such as ozone, pressure and radiation is quite limited. In general, SHAP analysis demonstrates that $PM_{2.5}$ is shaped by seasonal and transport-based determinants.

This framework is not only a benchmarking exercise but also provides a practical basis for short-term air-quality prediction in megacities. By incorporating locally measured

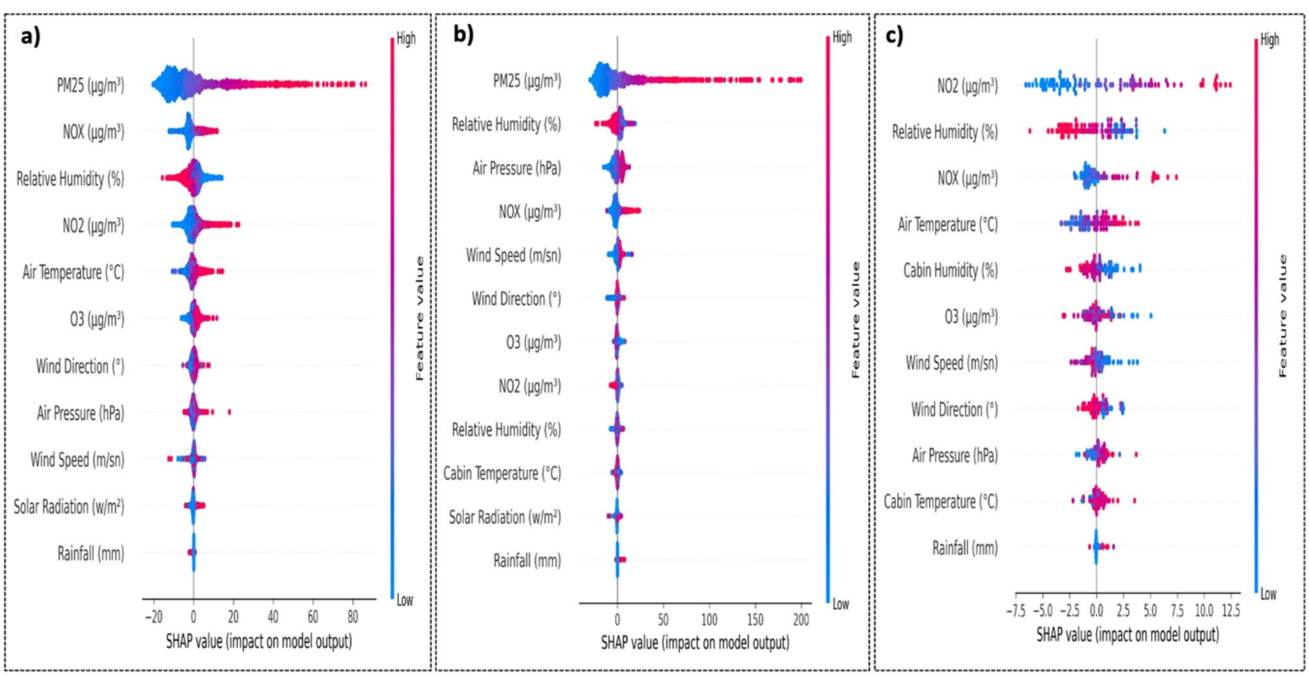

Fig. 11 Variable importance ranking obtained with SHAP for the prediction of PM_{10} concentrations during the winter period at **a** Arnavutköy (rural), **b** Bağcılar (urban), (**c**) Şile (urban background) AQMS

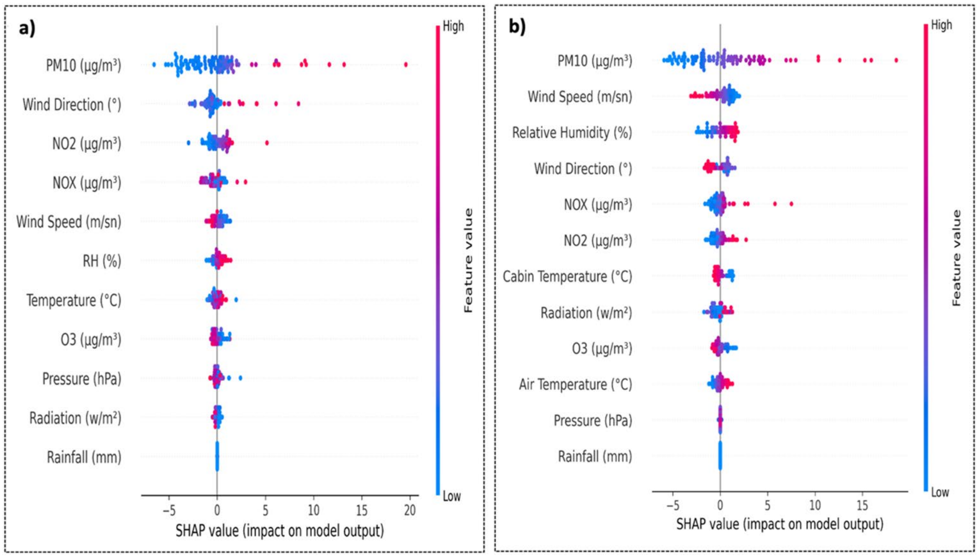

Fig. 12 Variable importance ranking obtained with SHAP for the prediction of $PM_{2.5}$ concentrations during the summer period at **a** Arnavutköy (rural) and **b** Bağcılar (urban) AQMS

meteorological variables, the models can be adapted into early-warning systems that inform residents and policy-makers of expected pollution episodes on an hourly to daily scale. While not intended as a full source-apportionment study, the SHAP analysis highlights the relative influence of meteorological and co-pollutant factors, offering insights into potential drivers of high-concentration events. Thus, the study demonstrates that machine-learning models can complement traditional chemical transport models by providing cost-efficient, site-specific, and season-aware estimations, which are valuable for public-health advisories and emission-control strategies. The SHAP-derived variable contributions were in line with established physical mechanisms of air pollution formation and dispersion, and with known atmospheric processes were observed. Additionally, extended SHAP analysis results are provided in the Supplementary Material (Figs. S17 and S18), which include variable importance rankings for all pollutant estimations across the three stations.

5 Limitations

This study has several limitations that should be taken into account when interpreting the results. First, the analysis is based on data from three air quality monitoring stations located within a single metropolitan area (Istanbul), which limits the spatial generalizability of the findings. Second, the modelling framework focuses exclusively on summer and winter periods; transitional seasons (spring and autumn), which may exhibit different emission patterns and meteorological dynamics, were not explicitly analyzed. Third, the models rely on co-pollutant measurements (e.g., PM_{10}, NO_2, NO_X) as input features, which may not be consistently available in all air-quality monitoring networks and could restrict the transferability of the framework to other operational settings. Finally, model performance and SHAP-based variable importance rankings are inherently context-dependent and may vary under different emission regimes, regulatory environments, or climatic conditions. Therefore, the results should be interpreted as site- and region-specific insights rather than universally applicable predictive relationships.

6 Conclusion

This work provides the initial city-wide comparison of seven ML algorithms for simultaneous prediction of PM_{10}, $PM_{2.5}$ and O_3 at three distinct monitoring sites in İstanbul. Also, to interpret the driving forces underlying the model predictions, SHAP (SHapley Additive exPlanations) analysis was performed at representative sites for the best performing PM_{10} and $PM_{2.5}$ models. The results confirmed the dominant role of co-pollutants (e.g. PM_{10} in $PM_{2.5}$ models) and key meteorological variables (such as wind speed, relative humidity and pressure), especially under winter conditions. Using 2021–2023 hourly observations, we found that ensemble-tree methods clearly dominate the predictive framework. For instance, in winter, ExtraTrees yielded the most accurate outcomes for PM_{10} both in Bağcılar and Arnavutköy, while the XGBoost outperformed its counterparts for $PM_{2.5}$ and O_3 at the same sites with highest predictive performance (e.g., RMSE values below 10 µg/m^3). Even at the semi-urban Şile station, where natural variability is larger, the best winter model (ExtraTrees) still explained 65% of PM_{10} variance.

Compared to the predictions made with regard to the winter conditions, summer predictions can be deemed as more challenging due to dispersed sources, stronger photochemistry and sea-breeze recirculation, which reduced predicted ability of the models most notably for PM_{10} in rural Arnavutköy. Nevertheless, the urban site still retained robust performance for three of the target variables, namely PM_{10}, $PM_{2.5}$ (by the MLP), and O_3 (by the XGBoost). These results confirm that secondary pollutants such as O_3 and $PM_{2.5}$ are easier to learn from joint pollutant-meteorology inputs than coarse PM, and season-specific modelling is essential to capture changing emission-dispersion regimes. The SHAP summary plots revealed that traffic-related indicators (NO_2, NO_x) and meteorological constraints (e.g. temperature and humidity) have a strong impact on PM variability in urban environments such as Bağcılar, while rural areas such as Arnavutköy exhibit a higher sensitivity to wind direction and long-range transport signatures.

From a practical perspective, coupling air-quality monitoring with on-site meteorological sensors proved critical insights., such as replacing same-station meteorology with nearest-neighbor data lowered average R^2 by ~0.08 (not shown). The study, therefore, recommends co-located measurements whenever possible and advocates hybrid deployment. In addition, it is evident that tree-based ensembles for winter-time prediction tasks and lightweight neural networks (MLP) for computationally efficient with respect to the predictions in dense urban cores, particularly under summer conditions. For further research, incorporating traffic flow, satellite AOD and atmospheric chemical transport outputs as additional features could also improve summer PM_{10} predictive ability and extend lead times beyond the one-hour horizon tested. The proposed framework demonstrates practical utility for short-term air-quality estimation and provides supporting evidence that can complement emission-control analyses in megacities such as Istanbul. Beyond retrospective evaluation, the proposed framework can be coupled with short-term meteorological predictions to support exploratory air-quality assessment at the monitoring-station scale. While not intended to replace process-based forecasting systems, the framework provides complementary, data-driven insights that may assist public-health interpretation and emission-management strategies when used alongside established physical models. At the same time, the interpretability analysis highlights the most influential meteorological and co-pollutant drivers, offering new insight into the mechanisms shaping air quality in urban and rural contexts.

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Data availability Air pollution data are publicly available online at https://havakalitesi.ibb.gov.tr/.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval No formal ethical approval is required.

Consent to participate All authors participated to the study.

Consent for publication The authors read and approved the final version of the manuscript.

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